

SolarTwin AI

Enterprise Solar Assessment Report

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Project Information

Parameter	Value
Location	Airoli, Airoli Station Road, Airoli Sector 2, Airoli, Navi Mumbai, Thane
Roof Area	17.02 m ²
Panels	5
Capacity	2.75 kW
Orientation	Portrait

Weather

Parameter	Value
Temperature	30.1 °C
Cloud Cover	98 %
Wind	17.6 km/h
Humidity	75 %

Solar Production

Parameter	Value
Annual Energy	4097 kWh
Daily Energy	11.2 kWh
Annual Saving	■ 32,775
Payback	5.0 Years
25 Year Profit	■ 654,378

AI Recommendations

- Large rooftop detected. Industrial-scale installation is feasible.
- Portrait layout provides the best packing efficiency.
- 2 rooftop obstacle(s) detected.
- Relocating obstacles could increase solar capacity.
- Frequent cloud cover may reduce annual generation.
- Suitable for residential self-consumption.

- Clean solar panels every 2–3 months for optimal efficiency.
- Perform annual electrical and structural inspections.

Executive Conclusion

SolarTwin AI concludes that the analysed rooftop is suitable for a photovoltaic installation of approximately 2.75 kW. The estimated annual production is 4097 kWh, with estimated annual savings of ■ 32,775. The generated layout maximizes the usable roof area while considering rooftop geometry and detected obstacles. The proposed design is recommended for further engineering validation before installation.